

Microsoft ADC Cybersecurity Skilling Program

Week 9 Assignment 16

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SECURE STORAGE, AND DATABASES

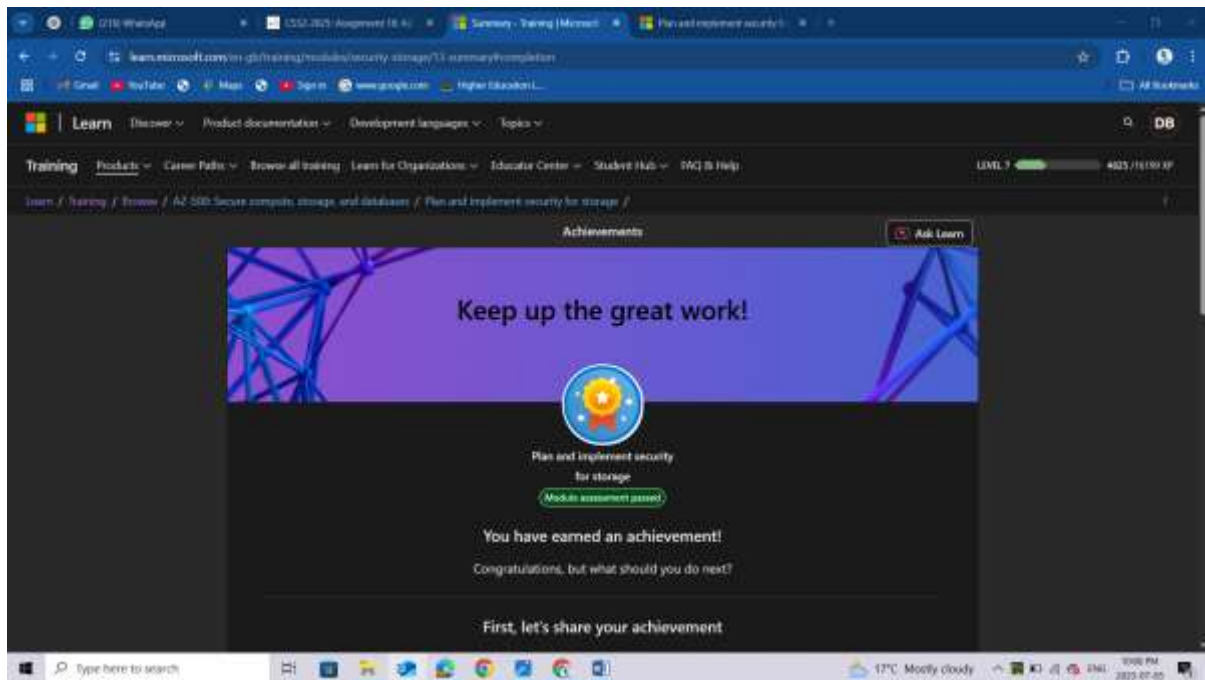
Introduction

In this module i will be focusing on two important modules that cover how to **plan and implement security for storage and Azure databases**. These modules are critical in understanding how to protect sensitive data stored in the cloud and ensure that only authorized users and services can access it. In the **storage security** module, I will learn how to secure Azure Storage accounts using features such as **shared access signatures (SAS)**, **service endpoints**, **private links**, **firewalls**, and **encryption mechanisms**. I will also explore how to manage identity-based access using **Azure Active Directory (Azure AD)**, **role-based access control (RBAC)**, and **storage account keys**. In the module covering **Azure SQL Database and Azure SQL Managed Instance**, I will focus on securing databases through **Transparent Data Encryption (TDE)**, **network security**, **auditing**, **Advanced Data Security**, and **threat detection**. These features help protect data both at rest and in transit, while also enabling proactive monitoring and response to potential security threats. By working through these modules, I aim to strengthen my ability to secure data resources in Azure environments, ensuring confidentiality, integrity, and compliance.

Tasks Completed

Plan and implement security for storage

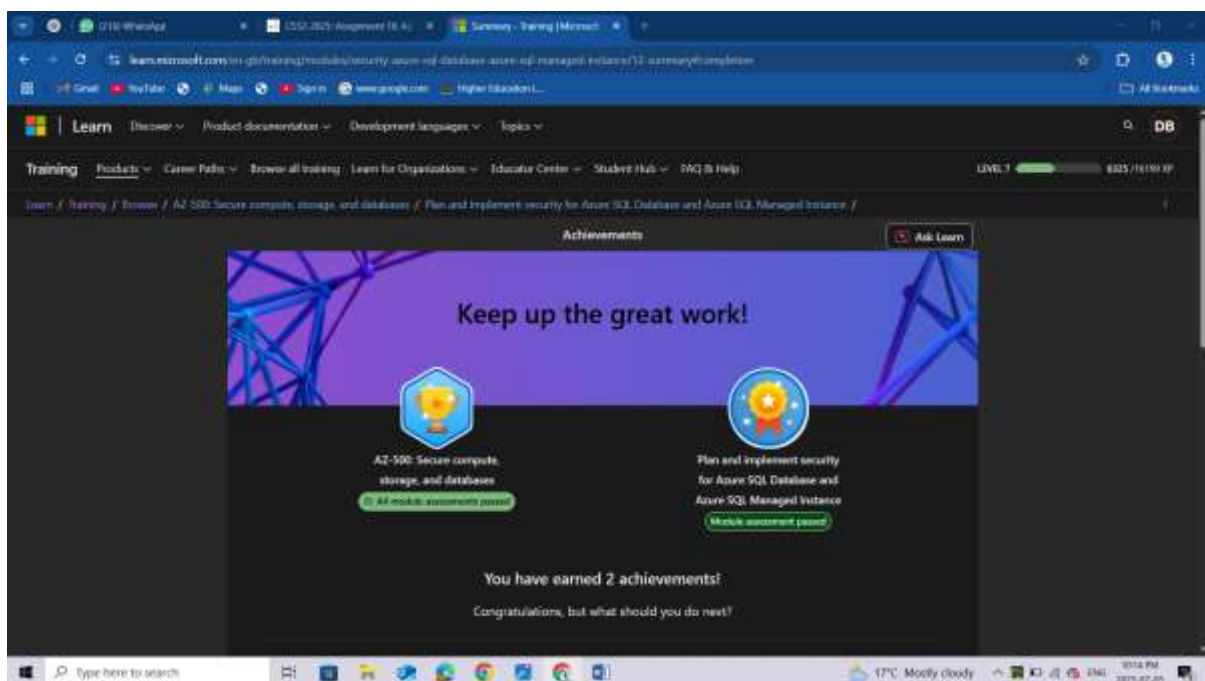
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Plan and implement security for Azure SQL Database and Azure SQL Managed Instance

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Conclusion

After completing the modules on securing Azure Storage and Azure SQL databases, I have developed a deeper understanding of how to effectively protect cloud-based data resources using Microsoft Azure's built-in security tools and best practices. In the **storage security**

module, I learned how to control access through **shared access signatures**, **storage account keys**, and **Azure RBAC**, while also using **network restrictions**, such as **firewall rules** and **private endpoints**, to isolate storage resources from unauthorized networks. Additionally, I explored how to apply **encryption at rest and in transit**, ensuring data remains protected throughout its lifecycle. In the **SQL security** module, I discovered how to implement key database protections such as **Transparent Data Encryption (TDE)**, **Always Encrypted**, and **network security controls** to restrict access to only trusted sources. I also worked with **auditing**, **Advanced Data Security**, and **threat detection** to enhance visibility and automate alerts on suspicious or risky activities within the database environment. Overall, these modules have equipped me with essential skills to design and implement a secure, compliant, and well-governed data infrastructure in Azure. I now feel more confident in my ability to secure both structured and unstructured data across different services in the cloud.